while on the mass m_2 it is μR_2 acting downwards. The separate equations for the motion of m_1 and m_2 are

$$m_1 g \sin\alpha - T - \mu R_1 = m_1 f \quad \ldots(1)$$

$$R_1 - m_1 g \cos\alpha = 0 \quad \ldots(2)$$

$$T - m_2 g \sin\beta - \mu R_2 = m_2 f \quad \ldots(3)$$

$$R_2 - m_2 g \cos\beta = 0 \quad \ldots(4)$$

Adding (1) and (3) and using (2) and (4), we get on solving

$$f = \frac{m_1(\sin\alpha - \mu\cos\alpha) - m_2(\sin\beta + \mu\cos\beta)}{m_1 + m_2} g \quad \ldots(5)$$

Eliminating f,

$$T = \frac{m_1 m_2 g}{m_1 + m_2}\left\{\sin\alpha + \sin\beta + \mu(\cos\beta - \cos\alpha)\right\} \quad \ldots(6)$$

1.16 Worked out examples :

Ex. 1. The velocity of a particle moving in a straight line is given by the relation $v^2 = ax^2 + 2bx + c$. Prove that its acceleration varies as the distance from a fixed point in the line.

Ans. Acceleration $= \dfrac{dv}{dt} = \dfrac{dv}{dx}\cdot\dfrac{dx}{dt} = v\dfrac{dv}{dx}$.

Now differentiating, $v^2 = ax^2 + 2bx + c$, w.r. to x we have

$$2v\frac{dv}{dx} = 2ax + 2b$$

$$\therefore \quad v\frac{dv}{dx} = a\left(x + \frac{b}{a}\right)$$

Thus the acceleration varies as the distance from the point $x = -\dfrac{b}{a}$.

Ex. 2. The law of motion in a straight line being $x = \dfrac{1}{2}vt$, show that the acceleration is constant.

Ans. Here $x = \dfrac{1}{2}vt$.

Differentiating both sides with respect to t, we have

or,

or,

or,

Integrating, we get,

where $-\log c$ is an inte

or,

Again, differentiatin

Hence acceleration

Ex. 3. A point mo
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velocity and f the ac

Ans. Here

or,

Differentiating wi

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Now,

$$\frac{dx}{dt} = \frac{1}{2}\left[v + t\frac{dv}{dt}\right]$$

or,
$$v = \frac{1}{2}v + \frac{1}{2}t\frac{dv}{dt}$$

or,
$$v = t\frac{dv}{dt}$$

or,
$$\frac{dv}{v} = \frac{dt}{t}$$

Integrating, we get, $\log v/c = \log t$

where $-\log c$ is an integration constant

or, $$v = ct$$

Again, differentiating with respect to t, we get

$$\frac{dv}{dt} = c = \text{constant}$$

Hence acceleration is constant.

Ex. 3. A point moves in a straight line so that its distance s from a fixed point at time t is proprotional to t^n. If v be the velocity and f the acceleration at any time t, show that

$$(n-1)v^2 = nfs$$

Ans. Here $s \propto t^n$

or, $s = kt^n$ [k = constant].

Differentiating with respect to t, we get

$$\frac{ds}{dt} = nkt^{n-1} \qquad \therefore v = nkt^{n-1}$$

Again, differentiating with respect to t, we have

$$\frac{d^2s}{dt^2} = n(n-1)kt^{n-1} \qquad \therefore f = n(n-1)kt^{n-2}$$

Now,
$$nfs = n^2(n-1)kt^{n-2}.kt^n$$
$$= (n-1)n^2k^2t^{2n-2}$$
$$= (n-1)\left(nkt^{n-1}\right)^2$$

$$= (n-1)v^2$$

Hence $$(n-1)v^2 = nfs$$

Ex. 4. A particle moves along a straight line according to the law $x^2 = at^2 + 2bt + c$. Prove that the acceleration varies as $1/x^3$.

Ans. We have $x^2 = at^2 + 2bt + c$.

Differentiating with respect to t, we obtain

$$2x\frac{dx}{dt} = 2at + 2b$$

or, $$x\frac{dx}{dt} = at + b$$

or, $$\frac{dx}{dt} = \frac{(at+b)}{\sqrt{at^2 + 2bt + c}}(at+b)(at^2+2bt+c)^{-\frac{1}{2}}$$

Again, diferentaiting with respect to t, we get

$$\frac{d^2x}{dt^2} = (at^2+2bt+c)^{-\frac{1}{2}}.a - (at+b)\frac{1}{2}2(at+b)(at^2+2bt+c)^{-\frac{3}{2}}$$

$$= \frac{(at^2+2bt+c)a - (at+b)^2}{(at^2+2bt+c)^{\frac{3}{2}}}$$

$$= \frac{ac-b^2}{(x^2)^{\frac{3}{2}}} = \frac{(ac-b^2)}{x^3}$$

Hence acceleration varies as $1/x^3$.

Ex. 5. The displacement x at time t of a particle moving along the x-axis is given by $t = \alpha x^2 + \beta x$, where α, β are positive constants. Express the velocity v of the p[illegible] at any moment as a function of x and prove that the re[illegible] of the particle is $2\alpha v^3$.

Ans. Here $$t = \alpha x^2 + [illegible]$$

$\therefore$ $$\frac{dt}{dx} = 2\alpha x [illegible]$$

or, $$v = \frac{dx}{dt} = \frac{[illegible]}{2\alpha x [illegible]}$$

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$n f s = k \cdot n^2 (n-1) t^{n-2} k \cdot t^n$

$= k^2 n^2 (n-1) \cdot t^{2n-2}$

$v = k n t^{n-1}$

$v^2 = k^2 \cdot n^2 t^{2n-2}$

$(n-1) v^2 = k^2 \cdot n^2 (n-1) \cdot t^{2n-2} = n f s$

Q3) A point moves in a straight line so that its distance s from a fixed point at time t is proportional to t^n. If v be the velocity ~~&~~ f is the acceleration at time t show that.

$(n-1)v^2 = nfs$.

$s = kt^n$

$\frac{ds}{dt} = knt^{n-1} = v$.

$f = \frac{d^2s}{dt^2} = kn(n-1)t^{n-2}$

straight

that

or,

Integrating with respect to t, we get

$$\frac{dx}{dt} = 15t - \frac{1}{6}t^3 + c$$

Initially, $t = 0$, $v = 0$. $\therefore$ $c = 0$

Hence, velocity $= \frac{dx}{dt} = 15t - \frac{1}{6}t^3$.

Again, integrating with respect to t, we get

$$x = \frac{15}{2}t^2 - \frac{1}{24}t^4 + d$$

Initially, $t = 0$, $x = 0$. $\therefore$ $d = 0$.

Hence, distance travelled is $x = \frac{15}{2}t^2 - \frac{1}{24}t^4$.

Ex. 7. A particle moves along a straight line so that after t secs its distance s from a fixed point O on the line is given by $s = (t-1)^2(t-2)$. Find its distance from O when its velocity is zero.

Ans. Here $s = (t-1)^2(t-2)$

$$\therefore \quad \frac{ds}{dt} = 2(t-1)(t-2) + (t-1)^2$$

$$= (t-1)(3t-5)$$

Hence $\frac{ds}{dt} = 0$, when $t = 1$ and $t = 5/3$.

At time $t = 5/3$, $$s = \left(\frac{5}{3}-1\right)^2\left(\frac{5}{3}-2\right)$$
$$= \frac{4}{9}\times\left(-\frac{1}{3}\right) = -\frac{4}{27}$$

At time $t = 1$, $s = 0$.

Hence the required distances from O are 4/27 and 0.

Ex. 8. The velocity of a particle moving in a straight line any instant t, when its distance from the origin is x, is give by $x = \frac{1}{2}v^2$; show that the acceleration of the particle is constan

Ans. Given, $x = \frac{1}{2}v^2$

Differentiating both sides w.r. to t, we get

$$\frac{dx}{dt} = v\frac{dv}{dt}$$

But for motion in a straight line, velocity $v = \frac{dx}{dt}$.

Hence
$$v = v\frac{dv}{dt}$$

or,

Hence the required work done

$$= f \times (x_2 - x_1)$$

$$= \frac{1}{2}\left(u_2^2 - u_1^2\right)$$

Ex. 10. A particle moves along a straight line according to the law $s^2 = 6t^2 + 4t^2 + 3$. Prove that the acceleration varies as $1/s^3$.

Ans. Given $s^2 = 6t^2 + 4t + 3$.

Differentiating w.r. to t, we have

$$2s\frac{ds}{dt} = 12t + 4$$

or,

$$\frac{ds}{dt} = \frac{6t+2}{\sqrt{6t^2 + 4t + 3}}$$

Differentiating again w.r. to t, we get

$$\frac{d^2s}{dt^2} = \frac{6}{\sqrt{6t^2+4t+3}} - \frac{2}{2}(6t+2)^2\left(6t^2+4t+3\right)^{-3/2}$$

$$= \frac{6\left(6t^2+4t+3\right) - (6t+2)^2}{\left(6t^2+4t+3\right)^{3/2}}$$

$$= \frac{18-4}{\left(s^2\right)^{3/2}} = \frac{14}{s^3} \propto \frac{1}{s^3}$$

Hence the result.

Ex. 11. Two particles of masses 12 lbs and 4 lbs are connected by a light inextensible string which passes over a small pulley. Find the velocity at the end of 2 secs and the space described in 2 secs.

Ans. Here $m_1 = 12$ lbs and $m_2 = 4$ lbs. Let f be the acceleration of the system. Using the formula $f = (m_1 - m_2)g/(m_1 + m_2)$, we get

$$f = \frac{12-4}{12+4} \times 32 = 16 \text{ ft/sec}^2$$

The apparent weight is greater than the true weight by fraction f/g.

(iv) When the lift moves downwards with acceleration $f (<g)$, the moving force

$$mg - R = mf \text{ or, } R = m(g-f)$$

Again, $$R = \frac{m(g-f)}{mg} . W = \left(1 - \frac{f}{g}\right) W$$

The apparent weight is less than the true weight by fraction f/g.

Note 1. When the downward acceleration of the lift is equal to the acceleraton due to gravity, $f = g$,

the moving force $= mg - R = mg$

$\therefore R = 0$, the apparent weight of the body on the thrust of the lift is zero.

Note 2. When the downward acceleration of the lift is greater than g, i.e., $f > g$, $R = m(g-f)$, is negative.

In this case, the floor of the lift will separate from the body unless there is a means of attachment of the body on the floor.

Note 3. If we place one body upon another and let the lower body fall down, the upper body will also fall with equal velocity and would exert any thrust on the lower body during the fall.

1.15 Motion of connected bodies :

We consider now the motion of two or more masses connected by light inextensible strings. We consider several simple problems of motion of connected systems.

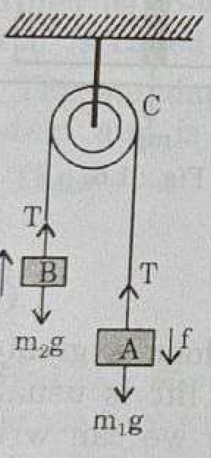

Fig. : 1.7

Case 1. *Two unequal masses m_1 and m_2 $(m_1 > m_2)$ are connected by a light inextensible string passing over a smooth fixed pulley. To find the resulting motion and the tension of the string.*

Let A and B be the two masses m_1 and m_2 respectively. They are connected by a light inextensible string that passes over a small smooth fixed pulley C.

Since the string is light and the pulley smooth, the tension of the string is the same throughout its length and it does not change in magnitude on either side of the pulley. Let T be the tension is absolute units.

Since $m_1 > m_2$, the mass m_1 will move downwards and m_2 will move upwards. As the string is inextensible, the downward displacement of m_1 at any time t will be equal to the upward displacement of m_2 at the same time t. Consequently, the downward velocity and acceleration of m_1 will be equal to upward velocity and acceleration of m_2 respectively.

Consider now the motion of mas m_1 only. The resultant force acting downwards is $m_1 g - T$.

The equation of motion is

$$m_1 g - T = m_1 f \qquad \ldots(1)$$

Similarly, the equation of motion of m_2 is

$$T - m_2 g = m_2 f \qquad \ldots(2)$$

To eliminate T, we add (1) and (2), to give

$$f = \frac{m_1 - m_2}{m_1 + m_2} g \qquad \ldots(3)$$

For eliminating f, we divide (1) by (2), to get

$$\frac{m_1 g - T}{T - m_2 g} = \frac{m_1}{m_2} \quad \text{or,} \quad T = \frac{2 m_1 m_2}{m_1 + m_2} g \qquad \ldots(4)$$

The tension T is the harmonic mean between the two weights $m_1 g$ and $m_2 g$.

The distance s described by either mass in time t starting from rest is given by

$$s = \frac{1}{2} f t^2 \quad \text{or,} \quad s = \frac{1}{2} \frac{m_1 - m_2}{m_1 + m_2} g t^2 \qquad \ldots(5)$$

Note 1. The tension T given by equation (4)is in a absolute units.

Note 2. The parts of the string which are not in contact with the pulley hang vertically and the force on the pulley = $2T$ is given by

$$2T = \frac{4 m_1 m_2}{m_1 + m_2} g$$

It is less than the sum of the weights of the masses.

$$\left[\text{Since } (\mathrm{m}_1 + m_2) g - \frac{4 m_1 m_2}{m_1 + m_2} g = \frac{(m_1 - m_2)^2}{m_1 + m_2} g > 0\right]$$

Note 3. If the pulley of weight W hangs by a hook from a nail, the force on the nail = $W + 2T$.

Ex. 11. Two particles of masses 12 lbs and 4 lbs are connected by a light inextensible string which passes over a small pulley. Find the velocity at the end of 2 secs and the space described in 2 secs.

Ans. Here $m_1 = 12$ lbs and $m_2 = 4$ lbs. Let f be the acceleration of the system. Using the formula $f = (m_1 - m_2)g/(m_1 + m_2)$, we get

$$f = \frac{12-4}{12+4} \times 32 = 16 \text{ ft/sec}^2$$

Let v be the velocity at the end of 2 secs and s be the space described at the end of 2 secs from rest.

$\therefore$

$$v = ft \ (\because \ u = 0) \text{ gives}$$
$$v = 16 \times 2 = 32 \text{ ft/sec}$$

and
$$s = \frac{1}{2} ft^2 = \frac{1}{2} \times 16 \times 4 = 32 \text{ ft}$$

Ex. 12. Two particles are connected by a light inextensible string passing over a smooth fixed pulley. If the system moves with an acceleration of 16 ft/sec^2, compare the masses of the particles.

Ans. Here $f = 16$ ft/sec^2. Let the masses of the particles be m_1 and m_2. Then, we have from the formula

$$f = \frac{m_1 - m_2}{m_1 + m_2} g$$

or,
$$16 = \frac{m_1 - m_2}{m_1 + m_2} \times 32 \quad \text{or,} \quad \frac{m_1 - m_2}{m_1 + m_2} = \frac{1}{2}$$

$\therefore$
$$m_1 : m_2 = 3 : 1$$

Ex. 13. Two masses 4lbs and $m$lbs are hung over a smooth ulley by means of a string. If the velocity of either mass in 2 cs is 16 ft/sec find m.

Ans. Let f be the accelertion of the mass m.

$\therefore \ v = u + ft$ gives $16 = f.2$ $[\because \ u = 0$ and $v = 16]$

$\therefore$
$$f = 8 \text{ ft/sec}^2$$

Using the formula,

$$f = \frac{m_1 - m_2}{m_1 + m_2} g$$

we get

$$8 = \left| \frac{4 - m}{4 + m} \right| \times 32$$

$[\because$ here $m_1 = 4$ and $m_2 = m]$

or,
$$(4 + m) = |4 - m| \times 4$$

$\therefore$ Either
$$4 + m = 4(4 - m)$$

or,
$$4 + m = 4(m - 4)$$

i.e.,
$$m = \frac{12}{5} \text{ lbs} \quad \text{or,} \quad m = \frac{20}{3} \text{ lbs}$$

Ex. 20. One end of a string is fixed; it then passes under a movable pulley to which a weight W is attached. The string then passes over a fixed pulley and a weight P is attached to its other end, all the three sections of the string being vertical. Show that neglecting the masses of the pulley the acceleration with which P descends is $(2P - W)g/(W + 4P)$ and the tension is $3\,WP/(W + 4P)$.

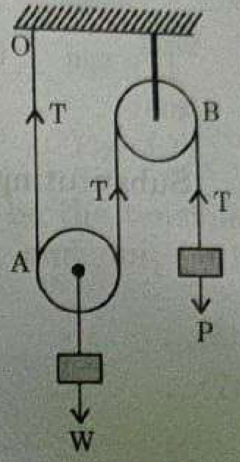

Fig. : 1.14

Ans. Let O be the fixed end of the string, B be the fixed pulley and A be the movable pulley. Now if the pulley A ascends a distance l, then the string on either side of the pulley becomes slack and slips over the pulley B so that the mass P descends through a distance $2l$. If f be the acceleration of A, then the acceleration of P is $2f$ downwards. Now, considering the motion of W and P separately, we have

$$2T - W = \frac{W}{g} f \qquad \ldots(1)$$

$$P - T = \frac{P}{g} 2f \qquad \ldots(2)$$

Multiplying (2) by 2 and adding to (1), we have

$$2P - W = (4P + W)\frac{f}{g}$$

$$\therefore \qquad f = \frac{2P - W}{W + 4P} \times g$$

Substituting this value of f in (1), we get

$$2T = W + \frac{W}{g}\left(\frac{2P - W}{W + 4P}\right)g$$

$$= \frac{W(W + 4P) + W(2P - W)}{W + 4P}$$

$$= \frac{6WP}{W + 4P}$$

$$\therefore \qquad T = \frac{3WP}{W + 4P}$$

Ex. 21. Two masses m_1 and m_2 $(m_1 > m_2)$ are connected by a light inextensible string passing over a light smooth pulley and are allowed to hang freely. Show that the acceleration of the centre of inertia downwards is

$$\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g$$

Ans. Let f be the acceleration of the system and T be th tension of the string. Then considering the motion of m_1 and m separately, we have

$$m_1 g - T = m_1 f \qquad \ldots(1)$$
$$T - m_2 g = m_2 f \qquad \ldots(2)$$

Adding (1) and (2), we get

$$(m_1 - m_2)g = (m_1 + m_2)f$$

$$\therefore \quad f = \frac{m_1 - m_2}{m_1 + m_2} \times g \qquad \ldots(3)$$

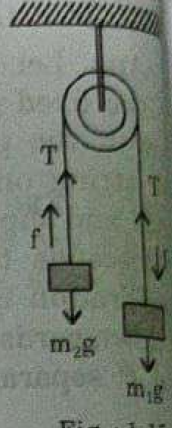

Fig. : 1.15

Substituting this value of f in (1), we have

$$T = m_1 g - \frac{m_1(m_1 - m_2)}{m_1 + m_2} g$$

$$= \frac{2m_1 m_2}{m_1 + m_2} g \qquad \ldots(4)$$

Now, we consider the motion of the centre of inertia of tw masses m_1 and m_2 i.e., of the combined mass $m_1 + m_2$. Let be the downward acceleration of the combined mass.

$$\therefore \quad (m_1 + m_2)g - 2T = (m_1 + m_2)f_1$$

or, $$f_1 = g - \frac{2T}{(m_1 + m_2)}$$

$$= g - \frac{4m_1 m_2}{(m_1 + m_2)^2} g \qquad \text{[using (4}$$

$$= g\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2$$

Ex. 22. Two particles of masses m_1 and m_2 are connected b a light inextensible string which passes over a smooth fixed pulle If $m_1 > m_2$, and m_1 descends with an acceleration f, show th the mass which must be taken from it so that it can ascend wi the same acceleration is $4m_1 fg/(f + g)^2$.

Ans. Let f be the acceleration of the system and T be th tension of the string. Considering the separate motion of m_1 a m_2, we have

$$m_1 f = m_1 g - T \qquad \ldots(1)$$
$$m_2 f = T - m_2 g \qquad \ldots(2)$$

Adding (1) and (2), we get